

System Modification Report

CDI / EDI Design Platform Security Update Log

Date: 2026-07-18

Version: 3.2.0-Secure

Executive Summary: This document logs all system updates performed on the CDI / EDI Design Platform. The user interface has been cleaned of all profile, card, and badges info. A secure Excel workbook logs login/logout history, user activities, and modifications.

1. Files Modified & Deleted

File Path	Action	Description of Change
server/app.js	MODIFIED	Mounted the secure log download route at GET /api/logs/download.
server/utils/excelHelper.js	MODIFIED	Upgraded sheet formatting for worksheets: Login_History (8 columns), User_Activity (5 columns), and Engineering_Modifications (7 columns).
client/src/components/Navbar.jsx	MODIFIED	Removed all user names, departments, welcome details, and role badges. Restored the "Download Login History" button calling /api/logs/download.
server/routes/authRoutes.js	MODIFIED	Connected login/logout history sheet updates, and registered client logging actions.
server/routes/designRoutes.js	MODIFIED	Integrated triggers for "Generate Design", "Generate P&ID", "Run Simulation", and modifications comparison.
server/routes/optimize.js	MODIFIED	Integrated triggers for "Optimize Design", "Run Simulation", and modifications comparison.
server/routes/equationsRoutes.js	MODIFIED	Logged Equation Editor events (creation, formula modification, deletion, library resets).
client/src/pages/EquationEditorPage.jsx	MODIFIED	Dispatched activity triggers on PDF/JSON exports and imports.

2. APIs Verified & Status Codes

Method	Endpoint	Status Code	Result
GET	/api/logs/download	200 OK	Securely downloads Login_History.xlsx workbook.
POST	/api/auth/login	200 OK	Logs login to spreadsheet, establishes cookie session.
POST	/api/auth/logout	200 OK	Logs logout, clears cookie, and destroys session.
GET	/api/equations	200 OK	Loads list of active equations.
POST	/api/equations	200 OK	Validates and appends custom equations or saves array.
PUT	/api/equations/:id	200 OK	Updates single equation definition.
DELETE	/api/equations/:id	200 OK	Deletes single equation and returns updated list.
GET	/api/design	200 OK	Retrieves design template defaults.
POST	/api/design	200 OK	Generates design sizing and calculation matrices.
POST	/api/optimize	200 OK	Optimizes CDI/EDI parameters via AI recommendations.

3. Excel Worksheet Architecture (Login_History.xlsx)

- **Login_History:** Stores attempts and logout timestamps. Columns: Login Time, Username, Status, IP Address, Browser, OS, Logout Time, Session Duration.
- **User_Activity:** Stores module actions. Columns: Date, Time, Username, Activity, Module.
- **Engineering_Modifications:** Stores engineering parameter changes. Columns: Date, Username, Module, Parameter, Old Value, New Value, Reason (if available).

4. Restoration of History Downloads

- **Download Login History Button:** Registered as a blue button next to the Equation Editor button in the Navbar. Clicking it securely queries `/api/logs/download` and downloads the spreadsheet directly.

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